COMSATS University Islamabad (Lahore Campus)

Assignment <3>

Course Title:	Design & Analysis of Algorithm	Course Code:	CSC301	Credit	3
Course Instructor/s:	Aqsa Nawaz	Program Name:	BCS		
Due Date:	**11:59 pm 10th May 2026**	**Maximum Marks:**		**20**	

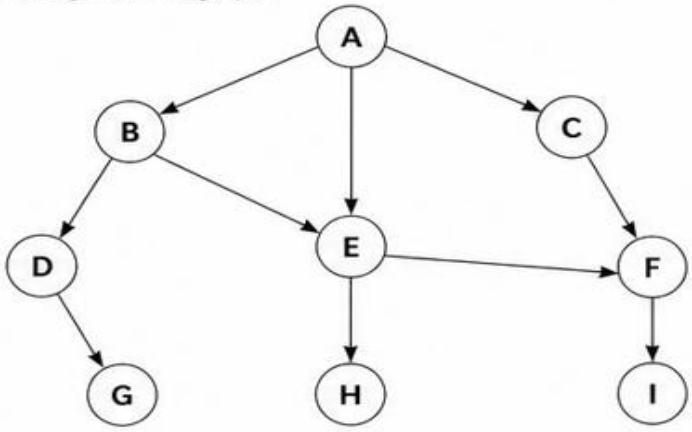
Submission Instructions:

- Submit In Pdf Format With File Named With Your Roll No like **FA24-BCS-000.pdf**
- DO NOT COPY YOUR WORK FROM ANY SOURCE (STRAIGHT ZERO MARKS!!.)
- Show **complete steps clearly** (no marks for only final answers).
- Draw diagrams neatly for BFS/DFS trees.
- Only Handwritten submissions are acceptable.
- Strictly No Late submissions allowed.

Marks: 20

Question 1: BFS

Consider the following directed graph.



a) Perform Breadth-First Search (BFS) starting from vertex A.
When multiple adjacent vertices are available, visit them in alphabetical order.

b) Show the step-by-step process in a table with the following columns:
Step No., Queue (front → rear), Node Dequeued, Newly Enqueued Nodes (in order), Visited Set

c) Write the BFS traversal order.

d) Produce the BFS tree and list its edges.

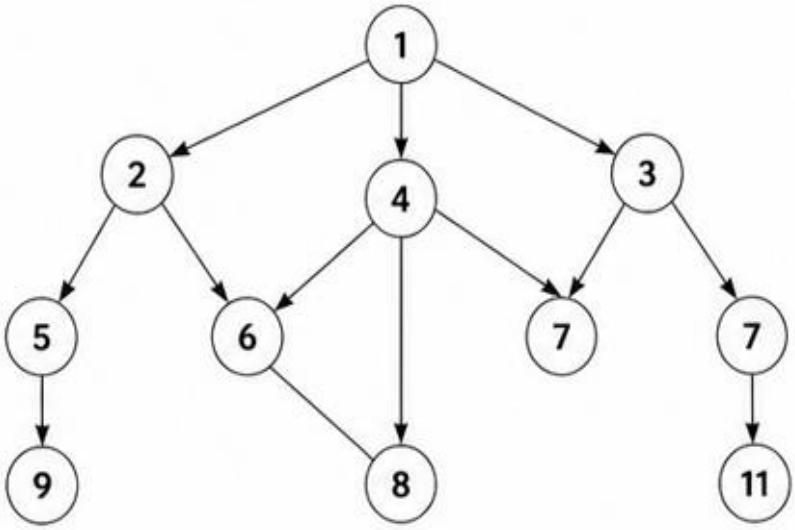
e) If the BFS is started from vertex B instead of A, what will be the BFS order?

f) What is the time and space complexity of BFS for this graph? Justify in terms of V and E.

g) Is it possible for BFS starting from A to visit I before H? Justify your answer.

Question 2: DFS

Consider the following directed graph.



a) Perform Depth-First Search (DFS) starting from vertex 1 using recursive approach.
When multiple unvisited adjacent vertices are available, visit the smallest numbered vertex first.

b) Show the step-by-step process in a table with the following columns:
Step No., Current Node, Action (Visit / Backtrack), Recursive Call Stack (top at right), Visited Set / Order

c) Write the DFS traversal order.

d) Classify all edges into: Tree Edges, Back Edges, Forward Edges, Cross Edges.
List all edges under each category based on DFS starting from 1.

e) If DFS is started from vertex 2 instead of 1, what will be the DFS order?

f) What is the time and space complexity of DFS? Justify in terms of V and E.

g) Give one practical real-world scenario where DFS is more useful than BFS and explain why.